Orthogonal terrain-following coordinates for the rational one-way Helmholtz equation: irregular terrain and bathymetry without staircases, angle limits or changes to the discrete nonlocal boundary condition

Mikhail S. Lytaev

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Abstract

Irregular terrain in tropospheric propagation and irregular bathymetry in underwater acoustics are usually represented in finite-difference parabolic equation (PE) solvers by staircases. We replace the staircase by a change of variables that maps the physical domain onto a rectangle. The map is required to (i) leave the Helmholtz operator in a form to which a rational (Padé, rational-interpolation) approximation of the one-way propagator of *arbitrary* order applies unchanged, so that no restriction on the propagation angle is introduced; and (ii) be the identity near the artificial boundary, so that the discrete nonlocal transparent boundary condition (NLBC) is used without modification. We show that sheared (Beilis–Tappert) coordinates cannot satisfy (i): the incident and reflected vertical wavenumbers at the boundary are not symmetric, so no function of the vertical operator reproduces the reflection. Conformal maps satisfy (i) but, by the identity theorem, cannot satisfy (ii). Orthogonal but non-conformal coordinates satisfy both. We construct them from the harmonic extension of the boundary, tapered to the identity, and from the orthogonal trajectories of the resulting family of level curves. In these coordinates the Helmholtz equation becomes a Sturm–Liouville problem with a generalised density $\tilde{\rho} = \rho h_\eta / h_\xi$ and a weight $m = (h_\eta/h_\xi)^2$. Each factor of the rational propagator then reduces to the density-weighted Numerov scheme with one extra diagonal weight. Range dependence of the metric is handled by a Liouville transformation and by a flux-conserving step $B_{n+1}^{-1/4} R B_n^{1/4}$. We write this step in a form that is the identity in the exterior, so the NLBC of the plain rational chain still applies. A discrete dispersion analysis shows that the transformed scheme is locally *equivalent* to the untransformed one on a grid with steps $h_\xi \Delta\xi$, $h_\eta \Delta\eta$ and reference wavenumber β/h_ξ. Hence the existing grid optimizer applies after a rescaling of its inputs, and the angle limit is the one of the chosen rational approximation, measured from the local marching direction. The method is implemented in JAX in `pywaveprop`. Against an exact solution for a beam reflected by a perfectly conducting slope (reflected beam up to 70°) its error is two to three orders of magnitude below that of the staircase. It agrees with a finely resolved RAM solution for a penetrable wedge. It is demonstrated on real terrain (SRTM-type elevation data) and real bathymetry and Argo sound speed profiles.

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1 Introduction

`pywaveprop` solves the one-way (pseudo-differential parabolic) Helmholtz equation with a rational approximation of the propagator,

$$\exp\big(i\beta\Delta x(\sqrt{1+L}-1)\big) \approx R(L) = \prod_{l=1}^{p} \frac{1+a_l L}{1+b_l L}, \tag{1}$$

a fourth-order (Numerov-type) vertical discretisation of L [1], an automated selection of Δx, Δz, β and of the order [2–4], and the discrete NLBC at the upper (troposphere) or lower (ocean bottom) artificial boundary [5]. Terrain and bathymetry are represented by a staircase: the field is set to zero below the terrain (or below the seabed) column by column. A staircase converges only at first order in Δz, it generates spurious diffraction at every step, it cannot represent a penetrable sloping interface with the correct continuity conditions, and for steep slopes it requires a vertical step much smaller than the one dictated by the propagation angle.

The aim of this note is a change of variables that maps the physical domain onto a rectangle, with the irregular boundary becoming a coordinate line, under three constraints:

1. no restriction on the maximum propagation angle is introduced: any angle that the rational approximation (1) resolves on flat ground is resolved over terrain, for an approximation of *arbitrary* order;

2. the upper discrete NLBC is used without modification;

3. the formulation covers both tropospheric radio-wave propagation (impenetrable terrain) and underwater acoustics (penetrable seabed with a density jump).

1.1 Related work

A literature survey (180 references, 42 of them available as PDFs; the list with DOIs is in `refs/README.md`) gives the following picture.

Shift maps. The Beilis–Tappert transformation $\zeta = z - T(x)$ [6], together with a phase factor $e^{ikT'\zeta + \frac{ik}{2}\int T'^2}$, maps the *narrow-angle* PE over terrain exactly onto the PE over flat ground with an additional potential $-2k^2\zeta T''$. It is the basis of TPEM, TEMPER and APM [7–9], of the acoustic PERUSE model [10] and of many later terrain PE codes [11–13]. In the Helmholtz equation, however, the shear produces the mixed derivative $-2T'\partial_x\partial_\zeta$, and the phase factor removes it only at the paraxial level. Wide-angle variants [8] split the square-root operator to first order in the slope. Published limits are about 10–20° for slopes, and the slope error is not reduced by raising the Padé order [13].

Rotations and mapped depth coordinates. The rotated PE [14, 15] is exact (the Laplacian is rotation invariant) and works with any Padé order, but only for piecewise-constant slopes; the field has to be interpolated at every slope break. Depth scalings $y = z/D(x)$ [16–19] and the curvilinear scheme of Janaswamy [20] are applied to an already paraxial equation, so each order needs a new derivation. Collins' energy-conserving [21] and single-scattering corrections improve the staircase interface conditions but keep the staircase.

Conformal and orthogonal maps. A conformal map $x+iz = F(\xi+i\eta)$ preserves the Laplacian, so the Helmholtz equation becomes the Helmholtz equation with index $n^2|F'|^2$. Dirichlet and Neumann conditions are invariant and the divergence form with density is preserved. Conformal maps were used for terrain and rough surfaces with *narrow-angle* split-step Fourier solvers [10, 22], for bent optical waveguides [23], and for water waves with wide-angle rational PEs [24]. The Earth-flattening transformation is the best-known example. Local orthogonal transformations combined with exact one-way (Dirichlet-to-Neumann) methods were developed by Lu and co-workers for bounded acoustic waveguides [25–27]; the equation they obtain has the same structure as ours, but no transparent boundary is involved.

Transparent boundaries. All exact discrete transparent boundary conditions [5, 28–30] assume an exterior whose coefficients do not depend on range. Kampanis [31, 32] used a terrain-fitting map that is the identity near the top boundary, so that the NLBC is unaffected, but with a narrow-angle finite-element PE; such maps are not conformal and produce mixed derivatives.

Gap. No published method combines a Helmholtz-exact map of the irregular boundary, an arbitrary-order rational propagator, and an exact discrete NLBC. The closest precedents combine two of the three: Earth flattening with [5], Kampanis's mapped finite elements with an NLBC, and Kirby's wide-angle PE in conformal coordinates. This note fills the gap.

1.2 Contributions and outline

Section 3 explains which maps are compatible with the rational propagator: sheared coordinates are not (proposition 1), orthogonal coordinates are (proposition 2), and conformal coordinates cannot be the identity near the NLBC (proposition 3). Section 4 constructs orthogonal coordinates that are the identity in the exterior. Section 5 derives the transformed one-way equation, including the Liouville and flux-conserving treatment of the range dependence of the metric. Section 6 gives the discretisation, section 7 the discrete dispersion analysis, section 8 the compatibility with the NLBC, section 9 the implementation, and sections 10 and 11 the validation and the real-data demonstrations.

2 Problem statement

We consider the two-dimensional Helmholtz equation for a time-harmonic field $\psi(x,z)$ (time dependence $e^{-i\omega t}$),

$$\rho\,\nabla\cdot\big(\rho^{-1}\nabla\psi\big)+k^2(x,z)\,\psi=0,\qquad k=\frac{\omega}{c}\,(1+i\eta_\alpha\alpha), \tag{2}$$

with $\rho\equiv 1$ in tropospheric propagation ($k^2=k_0^2n^2$, n including the Earth-flattening term of the modified refractivity M).

- **Troposphere.** z is the height above sea level, the domain is $\Omega=\{z>B(x)\}$, B is the terrain, and the ground is perfectly conducting: $\psi=0$ (horizontal polarisation) or $\partial_n\psi=0$ (vertical polarisation) on $z=B(x)$.
- **Ocean.** z is the depth, the sea surface $z=0$ is pressure release ($\psi=0$), and the seabed $z=B(x)$ separates the water (density ρ_w, sound speed $c_w(z)$) from a fluid bottom (ρ_b, c_b, attenuation α_b). On it $[\psi]=0$ and $[\rho^{-1}\partial_n\psi]=0$.

In both cases the field radiates through the outer side of the domain (up in the troposphere, down in the ocean bottom), where the medium does not depend on x and is homogeneous or has a linear $k^2(z)$ (standard atmosphere). There the domain is truncated by the discrete NLBC.

The rational one-way propagator. On flat ground, with $L=\beta^{-2}(\rho\partial_z\rho^{-1}\partial_z+k^2)-1$ and the envelope $u=\psi e^{-i\beta x}$, the outgoing part of ψ satisfies $u_x=i\beta(\sqrt{1+L}-1)u$, and one range step is (1). Each factor is a tridiagonal solve

$$b_l\,\delta_\rho^2 v+(1+\alpha\delta_\rho^2)\big[\tau^2(1+b_l\nu)v\big]=a_l\,\delta_\rho^2 u+(1+\alpha\delta_\rho^2)\big[\tau^2(1+a_l\nu)u\big], \tag{3}$$

with $\tau=\beta\Delta z$, $\alpha=1/12$, $\nu=k^2/\beta^2-1$ and the density-weighted second difference

$$\delta_\rho^2 u_j=\rho_j\Big(\frac{u_{j+1}-u_j}{\rho_{j+1/2}}-\frac{u_j-u_{j-1}}{\rho_{j-1/2}}\Big).$$

For $\rho\equiv 1$ this is the Numerov–Padé scheme of [1]. The discrete NLBC [5] expresses the last unknown of every factor through the history of the boundary values of all factors,

$$u^n_{N-1,l}-Q^{(0)}_{ll}u^n_{N,l}=\sum_{m=1}^{n}\sum_{l'}Q^{(m)}_{ll'}u^{n-m}_{N,l'}+\sum_{l'<l}Q^{(0)}_{ll'}u^n_{N,l'}, \tag{4}$$

where the kernels $Q^{(m)}$ are the inverse z-transform of a matrix function built from the coefficients a_l, b_l and from the discrete exterior equation. They depend only on the exterior scheme and on the chain of factors.

3 Which changes of variables are compatible with the rational propagator

A rational propagator of arbitrary order needs the one-way operator to be a function of a single vertical operator, $\exp(i\Delta\xi\sqrt{B})$. On flat ground with a Dirichlet boundary this is the case because the incident and reflected vertical wavenumbers are $\pm q$, the eigenfunctions $\sin q\eta$ of the Dirichlet operator ∂_η^2.

Proposition 1 (sheared coordinates)**.** *Let $\zeta=z-sx$ with constant $s=\tan\theta_s$, k constant, and $\psi=0$ on $\zeta=0$. The time-harmonic solutions $e^{ipx}f(\zeta)$ are $f=e^{iq_+\zeta}-e^{iq_-\zeta}$ with*

$$q_\pm=\frac{sp\pm\sqrt{(1+s^2)k^2-p^2}}{1+s^2},\qquad q_++q_-=\frac{2sp}{1+s^2}\neq 0. \tag{5}$$

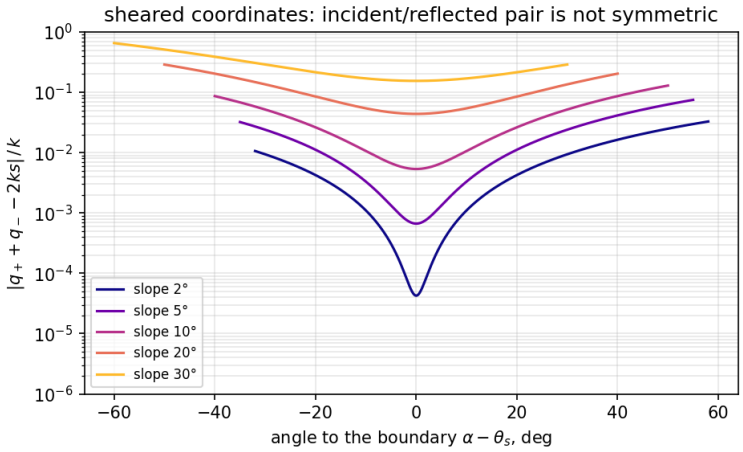


Figure 1: Asymmetry of the incident/reflected pair of vertical wavenumbers in sheared coordinates after the Beilis–Tappert gauge (proposition 1).

Hence for $s \neq 0$ the exact forward propagator in x does not commute with the Dirichlet realisation of ∂_ζ^2, and no propagator of the form $G(x)\,F(\partial_\zeta^2)$ with a multiplicative gauge G is exact for all p. The Beilis–Tappert gauge $G = e^{iks\zeta}$ symmetrises the pair only if $p = k(1+s^2)$, a value that propagating waves never reach since $p \leq k\sqrt{1+s^2}$. For a wave at the angle α to the x axis the remaining asymmetry is

$$\frac{q_+ + q_- - 2ks}{k} = 2s\big(\cos(\alpha - \theta_s)\cos\theta_s - 1\big).$$

Proof. Substituting $\psi = e^{ipx+iq\zeta}$ into $\psi_{xx} - 2s\psi_{x\zeta} + (1+s^2)\psi_{\zeta\zeta} + k^2\psi = 0$ gives $(1+s^2)q^2 - 2spq + p^2 - k^2 = 0$, whence (5). A wave $e^{i(k_x x + k_z z)}$ has $p = k_x + sk_z$; inserting $k_x = k\cos\alpha$, $k_z = k\sin\alpha$ gives the last formula. □

The asymmetry is $O(\theta_s^3)$ at grazing incidence on the slope, which is why the shift map works paraxially, and $O(\theta_s)$ otherwise (fig. 1). It does not decrease with the order of the rational approximation.

Proposition 2 (orthogonal coordinates)**.** *Let (ξ, η) be orthogonal coordinates with scale factors h_ξ, h_η and $r = h_\eta/h_\xi$. Then (2) reads*

$$\partial_\xi\Big(\frac{r}{\rho}\partial_\xi\psi\Big) + \partial_\eta\Big(\frac{1}{r\rho}\partial_\eta\psi\Big) + \frac{h_\xi h_\eta}{\rho}k^2\psi = 0, \tag{6}$$

without a mixed derivative. For frozen h_ξ, h_η the reflection $\eta \mapsto -\eta$ is a symmetry of (6). Hence the Dirichlet (Neumann) half-space solutions are odd (even) images, the incident and reflected wavenumbers are $\pm q$, and the exact forward propagator is $\exp\big(i\Delta\xi\sqrt{B}\big)$ with $B = h_\xi^2 k^2 + r^{-2}\partial_\eta^2$ realised with the boundary condition.

Proof. In orthogonal coordinates $\nabla\cdot(a\nabla\psi) = (h_\xi h_\eta)^{-1}\big[\partial_\xi(ah_\eta h_\xi^{-1}\partial_\xi\psi) + \partial_\eta(ah_\xi h_\eta^{-1}\partial_\eta\psi)\big]$; multiply by $h_\xi h_\eta/\rho$. With frozen coefficients the equation is $r\psi_{\xi\xi} + r^{-1}\psi_{\eta\eta} + h_\xi h_\eta k^2\psi = 0$, even in η. □

So orthogonality, not conformality, is what the rational propagator needs. A conformal map is the special case $r \equiv 1$. It cannot be used here because of the following fact.

Proposition 3 (conformal maps and the NLBC)**.** *Let F be analytic in a connected domain W and $F(w) = w + c$ on an open subset of W. Then $F(w) = w + c$ on all of W. Consequently a conformal map that straightens a non-flat boundary is not the identity (up to a shift) in any open set, in particular not near the NLBC; there $|F'|^2 \neq 1$, and the exterior index $n^2|F'|^2$ depends on range.*

Proof. Identity theorem for analytic functions. □

For a strip or periodic conformal map $|F'|^2 - 1$ decays like $e^{-2\pi\eta/\ell}$, where ℓ is the horizontal scale of the terrain. For a terrain feature 10 km long the residual is still $O(10^{-1})$ at a height of 1 km, and the NLBC kernel would see a range-dependent exterior. We therefore give up conformality and keep orthogonality.

4 Construction of the map

4.1 Family of level curves

The coordinate lines $\eta = \text{const}$ are the curves $z = Z(x, \eta)$ with

$$Z(x, \eta) = \eta + z_{\text{off}} + u(x, \eta), \tag{7}$$

where u is the harmonic extension of the boundary, tapered to zero. Let $b(x) = B(x) - z_{\text{off}} - \eta_b$ and $\hat{b}(\kappa)$ its Fourier transform. With $t = (\eta - \eta_b)/L$,

$$\hat{u}(\kappa, \eta) = \hat{b}(\kappa) \times \begin{cases} \eta/\eta_b \quad \left(\text{or } \dfrac{\sinh \kappa\eta}{\sinh \kappa\eta_b}\right), & 0 \le \eta < \eta_b \quad \text{(strip, ocean)}, \\ w(t)\, e^{-|\kappa|(\eta-\eta_b)}, & \eta \ge \eta_b, \end{cases} \tag{8}$$

with the C^2 taper $w(t) = 1 - t^3(10 - 15t + 6t^2)$ for $0 \le t \le 1$ and $w = 0$ for $t \ge 1$.

- **Troposphere (half-space map).** $\eta_b = 0$, $z_{\text{off}} = z_{\text{ref}}$. The terrain is the line $\eta = 0$ and the coordinates relax to $z = \eta + z_{\text{ref}}$.
- **Ocean (strip map).** $z_{\text{off}} = 0$, $\eta_b = \max B$. The surface $z = 0$ is $\eta = 0$, the seabed is $\eta = \eta_b$, and the coordinates are $z = \eta$ deep in the bottom.

The family has the following properties.

1. $Z(x, \eta_b) = B(x)$ exactly, and in the strip $Z(x, 0) = 0$.
2. *Identity region*: $Z = \eta + z_{\text{off}}$ for $\eta \ge \eta_\ell = \eta_b + L$.
3. Above the boundary, near it ($w \approx 1$), u is harmonic. A feature of horizontal scale ℓ decays over a height $\sim \ell$, so small-scale relief is confined near the boundary, as with a conformal map, and only the large-scale relief reaches the taper.
4. In the water column the default is the σ-coordinate $Z = \eta B(x)/\eta_b$, with $Z_\eta = B/\eta_b > 0$ for any bathymetry. The harmonic (sinh) profile keeps the near-conformal property, but the η-derivative of the strip kernel has negative lobes. Next to steep, high relief it then overshoots and makes $Z_\eta < 0$: on the Bermuda pedestal (section 11) this happens for every smoothing width up to 1 km.
5. *Regularity*: the map is invertible if $Z_\eta > 0$. In the water column of the strip map $Z_\eta = B/\eta_b > 0$. Above η_b write $u = w(t)U$, where U is the (Poisson) harmonic extension of b, so that
$$Z_\eta = 1 + w\, U_\eta + \frac{w'}{L}\, U.$$
   - With $z_{\text{ref}} = \max B$ (or $\eta_b = \max B$), $b \le 0$, hence $U \le 0$ by the maximum principle, and $w' \le 0$, so the taper term is non-negative and $Z_\eta \ge 1 - \max|U_\eta|$ for every L. Here $U_\eta = \mathcal{H}[b']$ at the boundary is a bound on the (smoothed) slopes.
   - With $z_{\text{ref}} = \overline{B}$ the taper compresses the high parts, and $Z_\eta \ge 1 - \max|U_\eta| - \frac{15}{8}\max(B - z_{\text{ref}})/L$.

The implementation checks $Z_\eta > 0$ (and $h_\xi > 0$) numerically and falls back to $z_{\mathrm{ref}} = \max B$.

The boundary is extended evenly at $x = 0$ and at the far end of a padded interval, and is filtered with a Gaussian of width σ (removing the corners of the piecewise-linear elevation samples). The even extension makes $u_x(0, \eta) = 0$, so the starting trajectory $\xi = 0$ is the vertical line $x = 0$ and the source aperture can be prescribed as a function of height, as on flat ground.

4.2 Orthogonal trajectories

Let $X(\xi, \eta)$ be the abscissa of the trajectory orthogonal to the level curves that leaves the identity region at $x = \xi$:

$$\frac{\partial X}{\partial \eta} = G(X, \eta) \equiv -\frac{Z_x Z_\eta}{1 + Z_x^2}, \qquad X(\xi, \eta_\ell) = \xi. \tag{9}$$

The tangent of the trajectory, $(X_\eta,\ Z_x X_\eta + Z_\eta)$, is orthogonal to the tangent $(1, Z_x)$ of the level curve: $X_\eta(1 + Z_x^2) + Z_x Z_\eta = 0$. The map is $(\xi, \eta) \mapsto (X, Z(X, \eta))$. Trajectories are integral curves of the smooth vector field $\nabla\eta$, which does not vanish because $Z_\eta > 0$, so they never cross. The scale factors are

$$h_\eta = \frac{Z_\eta}{\sqrt{1 + Z_x^2}}, \qquad h_\xi = P\sqrt{1 + Z_x^2}, \qquad P = \frac{\partial X}{\partial \xi}, \quad \frac{\partial P}{\partial \eta} = G_x(X, \eta)\, P, \quad P(\xi, \eta_\ell) = 1, \tag{10}$$

since $h_\eta^2 = X_\eta^2 + (Z_x X_\eta + Z_\eta)^2 = Z_\eta^2/(1 + Z_x^2)$ and $\partial_\xi(X, Z(X, \eta)) = P(1, Z_x)$. In the identity region $h_\xi = h_\eta = r = 1$. At the level η_b of the strip map the family is only continuous, so h_η and r jump while h_ξ stays continuous. This jump is harmless because the seabed is a density interface anyway (section 5).

Numerical construction. $\hat{b}$ is computed once by FFT on a uniform grid. For every level needed by the integrator, u, u_x, u_{xx}, u_η, $u_{x\eta}$ are evaluated by five inverse FFTs. The system (9)–(10) is integrated downwards from η_ℓ with RK4 for all columns simultaneously, on coarse levels with spacing about $\sigma/8$ placed separately on $[0, \eta_b]$ and $[\eta_b, \eta_\ell]$. The metrics are then interpolated to the computational nodes and half-nodes with cubic splines on each side of η_b. The orthogonality and the metric formulas were checked against finite differences of the computed (X, Z): the cosine of the angle between the coordinate lines is below $2 \cdot 10^{-4}$, at the level of the finite-difference error of the check.

Figure 2 shows the resulting coordinates for the examples used in this note.

5 The transformed one-way equation

5.1 Transformed Helmholtz equation and boundary conditions

The density depends only on η in computational coordinates, because the seabed is the line $\eta = \eta_b$. Equation (6) holds with:

- $\psi = 0$ at $\eta = 0$ (Dirichlet) or $\partial_\eta \psi = 0$ (Neumann, since $\partial_n = h_\eta^{-1}\partial_\eta$);
- $[\psi] = 0$ and $\big[(r\rho)^{-1}\partial_\eta \psi\big] = 0$ at $\eta = \eta_b$.

The latter is equivalent to $[\rho^{-1}\partial_n \psi] = 0$ because $(r\rho)^{-1}\partial_\eta \psi = h_\xi \rho^{-1}\partial_n \psi$ and h_ξ is continuous.

5.2 Liouville transformation

Range dependence of r enters through $\partial_\xi(r\rho^{-1}\partial_\xi)$. With $\psi = r^{-1/2}\varphi$, and using $\partial_\xi\big(r\partial_\xi(r^{-1/2}\varphi)\big) = r^{1/2}\varphi_{\xi\xi} - (r^{1/2})_{\xi\xi}\varphi$, (6) becomes

$$\varphi_{\xi\xi} + B\varphi = 0, \qquad B\varphi = \rho\, r^{-1/2}\partial_\eta\Big(\frac{1}{r\rho}\partial_\eta\big(r^{-1/2}\varphi\big)\Big) + \big(k^2 h_\xi^2 - V\big)\varphi, \qquad V = \frac{(r^{1/2})_{\xi\xi}}{r^{1/2}}. \tag{11}$$

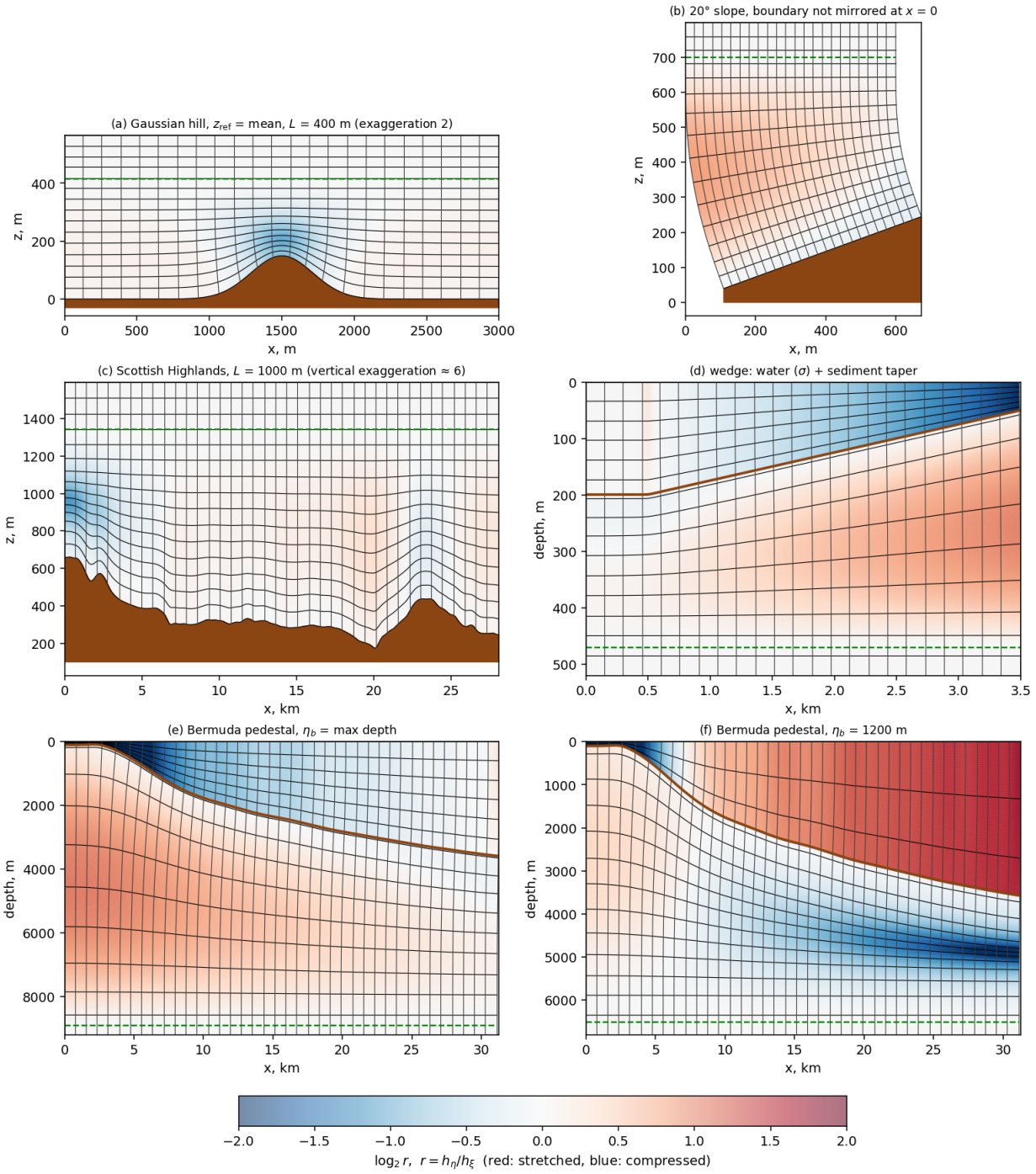


Figure 2: Orthogonal terrain-following coordinates on examples. Black lines: level curves η = const; grey lines: orthogonal trajectories ξ = const; colour: $\log_2 r$, $r = h_\eta/h_\xi$; green dashed line: top of the transition layer $\eta = \eta_\ell$, above which the map is the identity; brown line: seabed $\eta = \eta_b$. (a) Gaussian hill: small-scale relief decays over its own width. (b) 20° slope without the even extension at $x = 0$: the starting trajectory is curved. (c) Scottish Highlands (section 11). (d) Penetrable wedge: σ-coordinates in the water and the tapered extension in the sediment. (e,f) Bermuda pedestal with $\eta_b = \max B$ (strong compression of the shelf column) and with $\eta_b = 1200\,\mathrm{m}$ (compression moved into the sediment, expansion of the deep water). Panels (c)–(f) are vertically exaggerated, so the right angles appear distorted.

B is symmetric in $L^2(\rho^{-1}\mathrm{d}\eta)$. Its range dependence comes only from the metric and from the medium. V is a curvature-squared term, of relative size $(\lambda/\ell_\xi)^2$ with respect to k^2.

5.3 Flux-conserving one-way step

The one-way factorisation of (11) keeps the outgoing part, $\varphi_\xi = i\sqrt{B}\varphi - \frac{1}{4}B^{-1}B_\xi\varphi + \dots$. The second term is the transport (WKB) term. It expresses the conservation of the energy flux $\mathcal{F} = \mathrm{Im}\int \rho^{-1}\bar{\varphi}\varphi_\xi \,\mathrm{d}\eta = \|B^{1/4}\varphi\|^2$ instead of $\|\varphi\|^2$. In the standard PE in a slowly range-dependent medium it is neglected. Here it is not negligible: the coordinate lines rotate with height, so a beam changes its angle to the local marching direction, and the plain step $\varphi_{n+1} = \exp(i\Delta\xi\sqrt{B_{n+1/2}})\varphi_n$ preserves the wrong norm. In the slope tests of section 10.2 this costs 2–6 % of amplitude over a few hundred metres for beams reflected at 40–50°, in agreement with the WKB prediction $\big(\cos\theta_1/\cos\theta_2\big)^{1/2}$ ($\theta_{1,2}$ the angles of the beam to the coordinate lines at the two ends). We use the flux-conserving step

$$\varphi_{n+1} = B_{n+1}^{-1/4}\,\exp\big(i\Delta\xi\sqrt{B_{n+1/2}}\big)\,B_n^{1/4}\varphi_n, \tag{12}$$

which preserves $\mathcal{F}$ exactly when the propagator is unitary (cf. the energy-conserving PE [21, 33], where the same factors appear at a vertical interface). It is the essential new ingredient of the scheme; without it the transformation trades the staircase error for an amplitude drift.

Exterior-identity form. Since the middle propagator commutes with $B_{n+1/2}^{\pm1/4}$,

$$\varphi_{n+1} = \underbrace{\big[B_{n+1}^{-1/4}B_{n+1/2}^{1/4}\big]}_{C_2}\,\exp\big(i\Delta\xi\sqrt{B_{n+1/2}}\big)\,\underbrace{\big[B_{n+1/2}^{-1/4}B_n^{1/4}\big]}_{C_1}\,\varphi_n. \tag{13}$$

Each bracket is the identity wherever its two columns coincide, in particular in the whole exterior. The middle factor is the plain rational chain, so the NLBC (4) of the plain chain applies unchanged (section 8).

Rational approximation. With $B = \beta^2(1+L)$, the envelope $\varphi = e^{i\beta\xi}v$, and the diagonal Padé approximants $(1+L)^{\pm1/4} \approx \prod_{l=1}^{q}\frac{1+c_l^{\pm}L}{1+d_l^{\pm}L}$ ($c_l^- = d_l^+$, $d_l^- = c_l^+$; without the rotation introduced below, $q=1$ gives $(1+\frac{5}{8}L)/(1+\frac{3}{8}L)$),

$$v_{n+1} = \prod_{l=1}^{q}\frac{1+c_l^-L_{n+1}}{1+d_l^-L_{n+1}}\prod_{l=1}^{q}\frac{1+c_l^+L_{n+\frac12}}{1+d_l^+L_{n+\frac12}}\;R(L_{n+\frac12})\;\prod_{l=1}^{q}\frac{1+c_l^-L_{n+\frac12}}{1+d_l^-L_{n+\frac12}}\prod_{l=1}^{q}\frac{1+c_l^+L_n}{1+d_l^+L_n}\;v_n. \tag{14}$$

Rotated branch cut. $(1+L)^{1/4}$ has a branch cut on $L<-1$, and its Padé approximants have their poles and zeros on the cut, i.e. inside the evanescent part of the spectrum of the discrete L. The [2/2] approximant has a pole at $L=-1.38$ and a zero at $L=-1.18$. A pair $C_{1,2}$ at two slightly different columns is close to the identity, but for an eigenvalue next to a pole it is not, and it amplifies a spurious evanescent component. With the [2/2] approximant this produced a localised error of 14 % after 600 m in the 20° slope test of section 10.2. We therefore follow [34] and approximate

$$(1+L)^{1/4} = e^{i\gamma/4}\big(1+Y\big)^{1/4}, \qquad Y = (1+L)e^{-i\gamma} - 1,$$

by the $[q/q]$ Padé approximant in Y, with $\gamma = 45°$. Rewritten as factors in L, its poles and zeros lie at $L = (1-1/b)e^{i\gamma} - 1 \notin \mathbb{R}$, so $|1+bL|$ is bounded away from zero on the real spectrum. The constant prefactor cancels within each pair. Only the logarithmic derivative of the approximant enters $C_{1,2}$; for $q=2$ its relative error is below $5\cdot10^{-2}$ for $L\in[-0.5, 0.2]$ (propagation angles up to 45°). Since the correction itself is a few per cent of the amplitude, this gives errors of order 10^{-3}. With the rotated factors the 20° test is accurate to $1.2\cdot10^{-3}$.

5.4 Factors in terms of the physical field

All factors in (14) have the form $(1+cL)^{-1}(1+aL)$ with L taken at one column. They are solved for $\psi = r^{-1/2}\varphi$ with r of the same column, by the following identity.

Lemma 1. *Let $\varphi = r^{1/2}\psi$, $\tilde\rho = \rho r$, $m = r^2$ and $\nu = (k^2h_\xi^2 - V)/\beta^2 - 1$. Then*

$$r^{3/2}\beta^2(1+cL)\varphi = \mathcal{T}_c\psi \equiv c\,\tilde\rho\,\partial_\eta\big(\tilde\rho^{-1}\partial_\eta\psi\big) + \beta^2 m(1+c\nu)\psi.$$

Hence $(1+bL)\varphi_v = (1+aL)\varphi_u$ if and only if $\mathcal{T}_b\psi_v = \mathcal{T}_a\psi_u$.

Proof. $\beta^2(1+cL)\varphi = \beta^2(1-c)\varphi + cB\varphi$. Insert $B\varphi = \rho r^{-1/2}\partial_\eta\big((r\rho)^{-1}\partial_\eta\psi\big) + (k^2h_\xi^2 - V)r^{1/2}\psi$ and multiply by $r^{3/2}$. □

$\mathcal{T}_c$ is the operator of the flat density scheme (3) with the density ρ replaced by the generalised density $\tilde\rho = \rho h_\eta/h_\xi$ and with the extra weight m. Its natural interface conditions, $[\psi] = 0$ and $[\tilde\rho^{-1}\psi_\eta] = 0$, are the physical ones. Between columns the field is rescaled by $(r_{\rm old}/r_{\rm new})^{1/2} = (m_{\rm old}/m_{\rm new})^{1/4}$.

6 Discretisation

Vertical. On the uniform grid $\eta_j = j\Delta\eta$ every factor of (14) is

$$c\,\delta_{\tilde\rho}^2 v + (1+\alpha\delta_{\tilde\rho}^2)\big[\tau^2 m(1+c\nu)v\big] = a\,\delta_{\tilde\rho}^2 u + (1+\alpha\delta_{\tilde\rho}^2)\big[\tau^2 m(1+a\nu)u\big], \qquad \tau = \beta\Delta\eta, \tag{15}$$

a tridiagonal system. For $r \equiv 1$ it is exactly the existing scheme (3). For smooth $\tilde\rho$, with $g = \ln\tilde\rho$ and $D = \tilde\rho\partial_\eta\tilde\rho^{-1}\partial_\eta$, a Taylor expansion (checked symbolically) gives

$$\frac{\delta_{\tilde\rho}^2}{\Delta\eta^2} = D + \frac{\Delta\eta^2}{12}D^2 + \Delta\eta^2 E + O(\Delta\eta^4), \quad E\psi = \tfrac{1}{24}\big(g'^2 + g''\big)\psi'' + \tfrac{1}{24}\big(g''' + g'g'' - g'^3\big)\psi'.$$

The Numerov weight cancels the D^2 term. The scheme is therefore fourth-order in the wave part, with an additional second-order error of relative size $(\Delta\eta/\ell_\eta)^2/24$, where ℓ_η is the vertical scale of $\ln r$. In practice $\ell_\eta \gtrsim L \sim 10^2\lambda$, so this error is negligible.

Boundary and interface rows.

- *Dirichlet.* $v_0 = 0$ is eliminated from the system rather than imposed by a unit row: a unit row next to entries $\alpha\tau^2 m \gg 1$ is satisfied by a pivoting solver only up to amplified round-off.
- *Neumann.* Mirror node $v_{-1} = v_1$ with even coefficients.
- *Seabed node* $j_b = \eta_b/\Delta\eta$ (a grid node by construction). The equation is integrated over the cell, so $\tilde\rho_{j_b}$ is the harmonic mean of the one-sided values. m_{j_b} and ν_{j_b} are fixed by $m_{j_b}/\tilde\rho_{j_b} = \langle m/\tilde\rho\rangle$ and $m_{j_b}\nu_{j_b}/\tilde\rho_{j_b} = \langle m\nu/\tilde\rho\rangle$, where $\langle\cdot\rangle$ is the average of the two one-sided values.

Range. The coefficients $\nu, m, \tilde\rho$ (nodes) and $\tilde\rho$ (half-nodes) are tabulated on coarse columns ξ_c with spacing about $\sigma/4$ (the map varies on the scale of the smoothed boundary), and only below the identity region. They are interpolated linearly to ξ_n, $\xi_{n+1/2}$, ξ_{n+1}. The main chain is evaluated at the midpoint, which gives second order in the coefficient variation. The memory is $O(n_c\,\eta_\ell/\Delta\eta)$ instead of $O(n_x n_z)$.

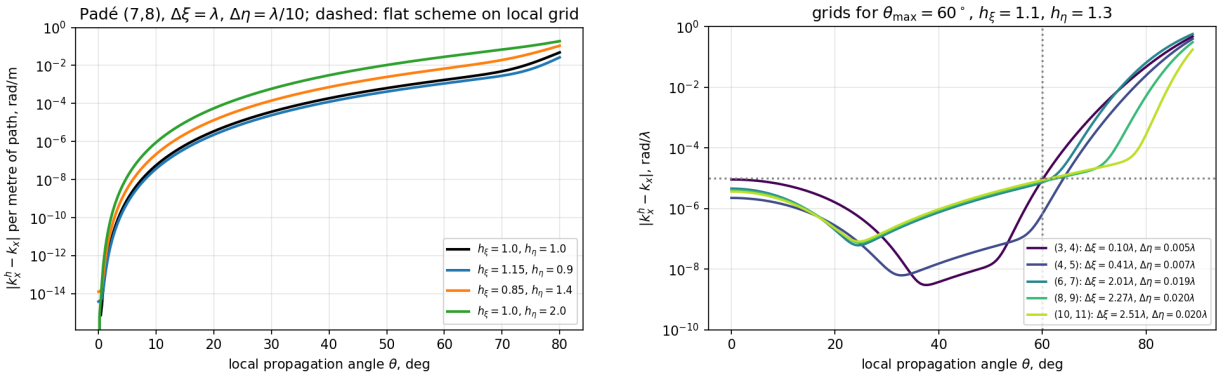


Figure 3: Left: dispersion error of the transformed scheme for several metric values on one grid (solid) and of the untransformed scheme on the local effective grid (dashed); the curves coincide to $4 \cdot 10^{-13}$ of the maximum error, as theorem 1 states. Right: Padé orders $(3,4)$–$(10,11)$ on the grids chosen for a 60° aperture in a strongly distorted cell ($h_\xi = 1.1$, $h_\eta = 1.3$); every order reaches the target accuracy up to the aperture.

7 Discrete dispersion analysis

Freeze h_ξ, h_η, ρ and k (so r, $m = r^2$, $\tilde\rho = \rho r$ and ν are constant), and consider an outgoing plane wave with physical angle θ to the local marching direction: $k_z = k\sin\theta$, computational vertical wavenumber $q = k_z h_\eta$, and $\psi_j^n = \lambda^n e^{iqj\Delta\eta}$. The flux factors of (14) cancel ($L_n = L_{n+1/2} = L_{n+1}$). With $s = \sin^2(q\Delta\eta/2)$, the symbol of (15) gives

$$L_h(q) = \nu - \frac{4}{\tau^2 r^2}\,\frac{s}{1-4\alpha s}, \qquad \lambda = R\big(L_h(q)\big), \tag{16}$$

while the exact amplification is $e^{i\Delta\xi(p-\beta)}$ with $p = h_\xi\sqrt{k^2 - k_z^2} = h_\xi k\cos\theta$. The dispersion error per unit physical length is

$$\varepsilon(\theta) = \frac{\left|\log\left(\lambda\, e^{-i\Delta\xi(p-\beta)}\right)\right|}{h_\xi\Delta\xi}. \tag{17}$$

Theorem 1 (local equivalence)**.** *$\varepsilon(\theta;\beta,\Delta\xi,\Delta\eta,h_\xi,h_\eta) = \varepsilon_0(\theta;\beta/h_\xi,\,h_\xi\Delta\xi,\,h_\eta\Delta\eta)$, where ε_0 is the dispersion error of the untransformed scheme* (3) *with reference wavenumber β/h_ξ and steps $h_\xi\Delta\xi$, $h_\eta\Delta\eta$.*

Proof. Let $\beta_e = \beta/h_\xi$, $\Delta x_e = h_\xi\Delta\xi$, $\Delta z_e = h_\eta\Delta\eta$.

- $\beta\Delta\xi = \beta_e\Delta x_e$, so R has the same coefficients: those of (1) depend on $\beta\Delta x$ only.
- $\nu = k^2h_\xi^2/\beta^2 - 1 = k^2/\beta_e^2 - 1$.
- $\tau r = \beta\Delta\eta\, h_\eta/h_\xi = \beta_e\Delta z_e$.
- $q\Delta\eta = k_z\Delta z_e$.

Hence L_h and λ coincide. The exact factor is $e^{i\Delta\xi(h_\xi k\cos\theta-\beta)} = e^{i\Delta x_e(k\cos\theta-\beta_e)}$, and the normalisations $h_\xi\Delta\xi = \Delta x_e$ agree. □

Figure 3 confirms theorem 1 numerically and shows its practical meaning.

Corollary 1 (angles)**.** *The transformation introduces no angle restriction. A wave is resolved if its angle to the* local *marching direction is within the aperture of the rational approximation on the local grid. Over terrain with slopes up to θ_s, waves reflected by the slopes reach $\theta_{\max} + 2\theta_s$ to the horizontal (and to the marching direction above the transition layer), so the aperture is set to $\min(\theta_{\max} + 2\theta_s, 85^\circ)$. This requirement is physical: it applies equally to a staircase, which usually ignores it.*

Corollary 2 (grid selection)**.** *Locally the scheme sees $k' = kh_\xi$, $k'_z = k_z h_\xi$, the vertical step $\Delta z' = r\Delta\eta$ for the operator $r^{-2}\partial_\eta^2$, and the range step $\Delta\xi$. With $h_\pm = \max / \min h_\xi$, the flat grid optimizer is applied with:*

- *the wavenumber interval $[k_{\min} h_-,\ k_{\max} h_+]$;*
- *the vertical wavenumber bound $k_z^{\max} h_-$, so that the smallest longitudinal wavenumber $h_-^2\big(k_{\min}^2 - (k_z^{\max})^2\big)$ is the pessimistic one;*
- *the error budget per unit computational length.*

The vertical step it returns resolves $k_z^{\max} h_-$. Since the scheme must resolve $k_z^{\max} h_+$ with the step $r\Delta\eta$, we take $\Delta\eta = \Delta z'\,(h_-/h_+)/\max r$. The stability properties of the flat scheme (spectrum of L_h in $(-\infty, \nu]$, $|R| \le 1$ there) carry over, because L_h is the flat-scheme symbol at the local parameters.

Remark 1 (layered media). In computational coordinates the seabed is a horizontal interface, across which the longitudinal wavenumber p is conserved. For an aperture $\theta_{\max}$ defined in the water, the field therefore has $p \ge \min_{\text{water}} k \cos\theta_{\max}$ in the bottom too. The lower end of the wavenumber interval is $\min_{\text{water}} k$, not the smaller bottom wavenumber; using the latter would forbid apertures beyond the bottom critical-angle complement ($\approx 64°$ for 1530/1700 m/s), although on a downslope each bottom reflection lowers a ray's angle by $2\theta_s$, so energy beyond the critical angle can return to long-range paths.

The remaining errors, which the frozen-coefficient analysis does not see, are:

1. the coefficient-variation truncation, $O\big((\Delta\eta/\ell_\eta)^2\big)$;
2. the linear interpolation of the coefficient tables in range, $O\big((\Delta\xi_c/\ell_\xi)^2\big)$;
3. the one-way (no backscatter) approximation of the range-dependent operator, which is the same approximation as for any range-dependent medium;
4. the error of the $[q/q]$ approximation of $(1+L)^{\pm 1/4}$, which acts on a quantity that is already of first order in $\Delta\xi\,\partial_\xi \ln B$.

The validation in section 10.2 bounds them all together: there the field is accurate to $1.2 \cdot 10^{-3}$ for a beam reflected at 50°, and to $3 \cdot 10^{-3}$ at 70°.

8 Compatibility with the upper nonlocal boundary condition

Theorem 2. *Let the computational domain be $0 \le \eta \le \eta_{\max}$ with $\eta_{\max} \ge \eta_\ell + 2\Delta\eta$, and let the medium in the exterior be range-independent. Then the discrete NLBC* (4) *for the chain $R = \prod_l (1 + a_l L)/(1 + b_l L)$, computed for the untransformed exterior scheme, is the exact discrete transparent boundary condition of the scheme* (14)–(15)*.*

Proof. 1. In the rows $j \ge \eta_\ell/\Delta\eta$ the map is the identity: $h_\xi = h_\eta = r = m = 1$, $V = 0$ and $\tilde\rho = \rho$. Since the taper is C^2, these values hold exactly at the nodes and half-nodes used by the last rows, so the scheme (15) in these rows coincides with (3) with $z = \eta + z_{\text{off}}$.

2. The flux factors are applied as the pairs C_1, C_2 of (13). Each pair is the identity operator in the exterior because its two columns coincide there. Numerically the pairs are solved on the rows below η_ℓ plus a margin of four wavelengths, with the top value passed through, and the rows above are left untouched.

3. What reaches the boundary is therefore produced by the chain R alone, with exterior coefficients independent of n. The derivation of [5] (discrete z-transform in range of the exterior recurrences of the chain) applies verbatim, including the linear-ν (Airy) exterior.

□

Remark 2. Writing the flux step as (12) with the factors placed around R would also be correct. The NLBC would then have to be computed for a chain of $p + 2q$ factors containing reciprocal pairs. The matrix function in the inverse z-transform then becomes defective, and the adaptive quadrature of the kernels does not converge, which we observed. The form (13) avoids this, and the NLBC kernels are the same as on flat ground.

In the ocean the "upper" boundary is the bottom of the domain, deep in the homogeneous sediment. The identity region is reached at depth $\eta_\ell = \eta_b + L$, so the NLBC sees the homogeneous bottom.

9 Implementation in `pywaveprop`

- `pywaveprop/terrain_transform.py`: `OrthogonalTerrainMap` (level curves (7)–(8), trajectories and metrics (9)–(10); NumPy).
- `pywaveprop/helmholtz_tf_jax.py`: `TerrainFollowingHelmholtzPropagator`, a subclass of `RationalHelmholtzPropagator`. It reuses the Padé and rational-interpolation coefficients, the grid optimizer (corollary 2) and the NLBC kernels, and replaces the vertical operator by (15) and the range step by (14). The march is a `jax.lax.fori_loop`.
- Front-ends: the option `terrain_method='transform'` of `RWPComputationalParams` (troposphere; Dirichlet or Neumann ground for H or V polarisation) and of `UWAComputationalParams` (penetrable seabed following the bathymetry). The results carry the physical coordinates of the output nodes.

Algorithm 1 One range step of the terrain-following rational propagator

1: interpolate $(\nu, m, \tilde{\rho}, \tilde{\rho}_{1/2})$ at ξ_n, $\xi_{n+1/2}$, ξ_{n+1}
2: $\psi \leftarrow C_1\psi$: q factors $\mathcal{T}_{d^+}^{-1}\mathcal{T}_{c^+}$ at ξ_n; rescale by $(m_n/m_{n+1/2})^{1/4}$; q factors $\mathcal{T}_{d^-}^{-1}\mathcal{T}_{c^-}$ at $\xi_{n+1/2}$ ▷ rows $< \eta_\ell$+margin
3: **for** $l = 1, \ldots, p$ **do**
4: solve $\mathcal{T}_{b_l}\psi' = \mathcal{T}_{a_l}\psi$ at $\xi_{n+1/2}$ with the NLBC row (4); $\psi \leftarrow \psi'$
5: **end for**
6: $\psi \leftarrow C_2\psi$ (as in line 2 with $\xi_{n+1/2} \to \xi_{n+1}$)

Cost. A step costs $p + 4q$ tridiagonal solves. The $4q$ flux solves are restricted to the transition layer. The coefficient tables add $O(n_c\eta_\ell/\Delta\eta)$ memory. The map is built once, in seconds.

A bug found on the way. The flat-ground comparison of the new propagator with the existing one exposed a defect in `RationalHelmholtzPropagator`. After the pytree round trip performed by `jax.jit`, the zero placeholder of the lower-NLBC kernels switched the lower NLBC on. The Dirichlet row then read $u_1 = 0$, so the ground was effectively one grid step above $z = 0$. This is fixed in this work. With the fix the two propagators agree to $4 \cdot 10^{-14}$ over flat ground.

10 Validation

10.1 Flat ground

Over flat ground the map is the identity and the terrain-following propagator reproduces `RationalHelmholtzPropagator`, including the NLBC, to $4 \cdot 10^{-14}$ (unit test `TestFlatGroundEquivalence`).

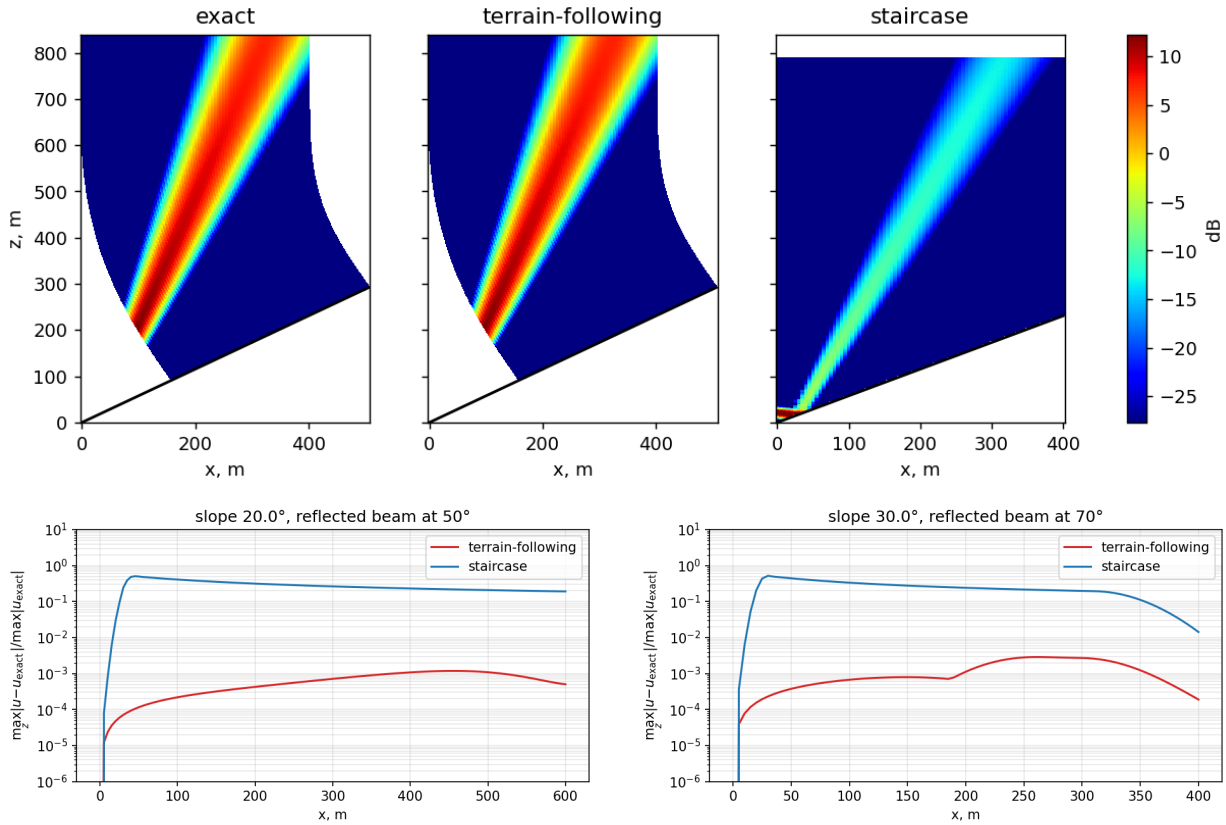


Figure 4: Beam reflected by a perfectly conducting slope. Top: $|u|$ for $\theta_s = 30°$ (reflected beam at 70°); the terrain-following domain is bounded by the curved trajectories $\xi = 0$ and $\xi = x_{\max}$. Bottom: error versus range for $\theta_s = 20°$ and 30°.

10.2 Exact solution: beam reflected by a perfectly conducting slope

The incident field is a complex-source-point beam at 300 MHz, an exact solution of the Helmholtz equation, with waist at $(-50, 30)$ m, Rayleigh parameter 100 m and direction $-10°$. Its image in the line $z = x \tan\theta_s$ gives the exact field above the Dirichlet slope. The reflected beam leaves at $2\theta_s + 10°$, i.e. 30, 50 and 70° for $\theta_s = 10$, 20 and 30°.

- Both solvers start from the exact field: on the curved trajectory $\xi = 0$ (boundary not mirrored at $x = 0$) and on the line $x = 0$ respectively.
- Both use Padé $(7, 8)$, the aperture $\theta_{\max} = 10° + 2\theta_s + 5°$ and grids chosen by the respective optimizers.
- The error is $\max|u - u_{\text{exact}}| / \max|u_{\text{exact}}|$ over the computational domain above the ground.

Table 1: Exact slope test (Padé $(7, 8)$, precision parameter 10^{-2}).

		terrain-following			staircase		
θ_s	aperture	Δx, m	Δz, m	error	Δx, m	Δz, m	error
10°	35°	2.50	0.0345	$1.4 \cdot 10^{-4}$	2.50	0.0625	$5.4 \cdot 10^{-1}$
20°	55°	2.50	0.0192	$1.2 \cdot 10^{-3}$	2.50	0.0345	$6.9 \cdot 10^{-1}$
30°	75°	1.00	0.0093	$2.9 \cdot 10^{-3}$	1.00	0.0159	$8.1 \cdot 10^{-1}$

The terrain-following error (table 1, fig. 4) is 10^{-4}–$3 \cdot 10^{-3}$ up to a reflected beam at 70°, on grids comparable to those of the staircase. The staircase loses most of the reflected beam, and its error

is 0.5–0.8. Refining the staircase grid for $\theta_s = 20°$ to $\Delta z = 0.005\,\mathrm{m}$ ($\lambda/200$) leaves the error at 0.54, and with $\Delta x = 0.625\,\mathrm{m}$ it is still 0.45: the staircase error is geometric and decreases only slowly.

The flux-conserving step is essential:

- With the Liouville rescaling alone ($q = 0$) the error in the 20° case is $6.4 \cdot 10^{-2}$, an amplitude drift along the reflected beam. Without the rescaling it is larger still.
- With unrotated [2/2] factors a spurious evanescent component grows to 0.14 (section 5).
- The rotated factors give $1.2 \cdot 10^{-3}$.

10.3 Penetrable wedge

The upslope wedge of the ASA benchmark [35] has the following parameters:

- 25 Hz, water $c = 1500\,\mathrm{m/s}$;
- bottom 1700 m/s, $\rho = 1.5$, $0.5\,\mathrm{dB}/\lambda$;
- source at 100 m;
- depth 200 m with a flat lead-in to 500 m, then a 2.86° upslope to 3.5 km.

All solvers use the same smoothed bathymetry ($\sigma = 20\,\mathrm{m}$) and start from the self-starter of RAM 1.5 at $r = 1\,\mathrm{m}$; a patched copy of RAM writes it out. RAM is single precision. On fine grids its tridiagonal systems lose all accuracy: at $\Delta r = 1\,\mathrm{m}$, $\Delta z = 0.05\,\mathrm{m}$ the field is wrong by 77 dB. We therefore rebuilt it in double precision. It then converges: the grids $(2.5, 0.1)\,\mathrm{m}$ and $(1, 0.025)\,\mathrm{m}$ differ by 0.013 dB (median), and the latter is the reference. The terrain-following

Table 2: Penetrable wedge: median / 90th percentile of $|\Delta\mathrm{TL}|$ (dB) with respect to the RAM reference, $r > 200\,\mathrm{m}$.

solver (grid)	$z = 30\,\mathrm{m}$	$z = 150\,\mathrm{m}$
terrain-following ($\Delta x = 5$, $\Delta z = 0.5\,\mathrm{m}$)	0.05 / 0.22	0.13 / 0.33
same scheme, stair-step bottom ($\Delta x = 5$, $\Delta z = 1.0\,\mathrm{m}$)	0.54 / 1.29	0.82 / 2.07
RAM ($\Delta r = 10$, $\Delta z = 0.5\,\mathrm{m}$)	1.66 / 6.70	2.79 / 10.42

solution agrees with the reference to 0.05–0.13 dB (table 2, fig. 5). The stair-step bottom in the same scheme gives 0.5–0.8 dB.

Refining Δz from 2 to 0.25 m does not reduce either error (terrain-following 0.09–0.09 dB, stair-step 0.68–0.71 dB), and neither does refining Δx to 1.25 m or raising the flux order to $q = 3$. The two floors have different origins:

- *Stair-step.* This is the energy-conservation defect of one-way models across vertical density steps [33]. Each step makes an error $O(\Delta h)$ and there are $O(1/\Delta h)$ steps, so the bias does not vanish. RAM avoids it through its density-scaled variable.
- *Terrain-following.* The residual of about 0.1 dB is the difference between two one-way models of the same sloping interface (RAM's stair-steps, ours continuous). It is grid-independent. It is larger at the receiver that lies in the sediment beyond 1.5 km.

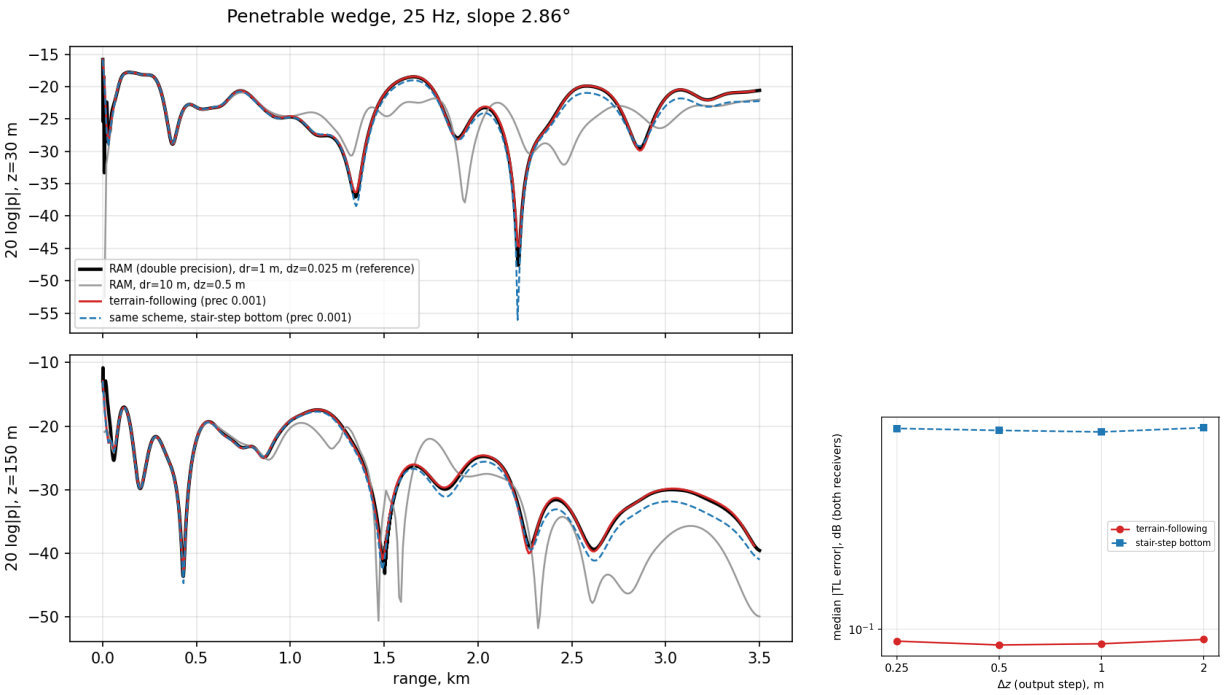


Figure 5: Penetrable wedge. Left: $20 \log |p|$ at 30 m and 150 m depth. Right: median error versus Δz at the same range step.

11 Real-world demonstrations

11.1 Troposphere: Scottish Highlands, GFS refractivity

The path runs from 57.2163°N 4.9214°W (top of a 664 m ridge) to 57.40°N 4.60°W (28 km). The data are:

- elevation tiles of `pywaveprop.environment` at 0.0005° resolution (terrain from 166 to 664 m, smoothed with $\sigma = 100$ m, maximum slope 11.1°);
- the modified refractivity of the GFS analysis 2026-10-09T12:00Z interpolated along the path and averaged over range (fig. 6, right).

A 300 MHz H-polarised antenna on a 30 m mast (beam 4°, tilted 1° down) radiates over perfectly conducting ground. The aperture is $5° + 2 \cdot 11.1° = 27.3°$ (corollary 1) for both solvers, and the staircase solver sees the same smoothed terrain. Path loss 10 m above the ground (median / 90th percentile of the difference):

- the two precision settings of the terrain-following solver differ by 0.08 / 0.17 dB;
- those of the staircase solver differ by 0.15 / 0.48 dB;
- the two solvers at the finer setting differ by 1.09 / 3.04 dB (at 50 m: 0.44 / 1.35 dB).

Each solver is thus converged with respect to its own grid; the two agree to about
SI1dB (median)
SI10m above the ground and
SI0.4dB at
SI50m. The largest differences, up to a few decibels, occur on the slopes facing the transmitter (21–23 km and the first kilometres). There the staircase solution carries small-scale ripple that the smooth-boundary solution does not, consistent with scattering by the steps (section 10.2). Run times: terrain-following 468 and 1390 s, staircase 73 and 117 s.

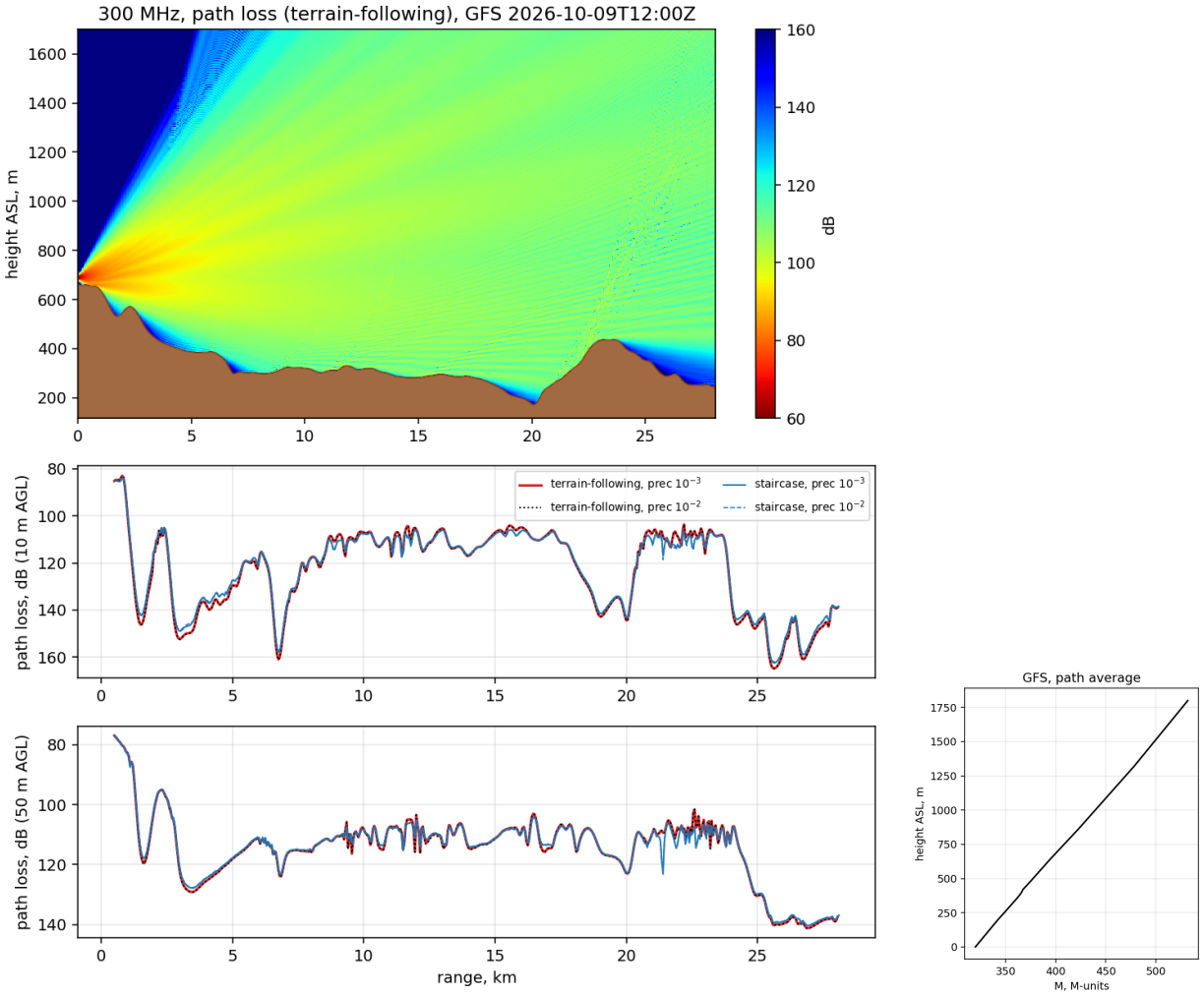


Figure 6: Scottish Highlands, 300 MHz. Path loss of the terrain-following solver (top), path loss 10 m and 50 m above the ground for both solvers and two precision settings, and the GFS M-profile.

11.2 Underwater acoustics: Bermuda pedestal, Argo sound speed

The path runs from 32.475°N 64.598°W (100 m deep, on the edge of the Bermuda platform) to 32.70°N 64.40°W (3400 m). The data are:

- bathymetry from the elevation tiles at 0.002° (smoothed with $\sigma = 500$ m, maximum slope 16.1°);
- the mean Argo sound speed profile of the area (2020–2023) to 2000 m, extended with the deep isothermal gradient 0.0165 s^{-1};
- a fluid bottom (1700 m/s, 1.5, 0.5 dB/λ) whose interface follows the bathymetry.

A 100 Hz source 50 m deep radiates down the slope. All solvers start from RAM's self-starter. The terrain-following domain extends to 9063 m, and the NLBC lies in the homogeneous bottom. The reference is double-precision RAM at $\Delta r = 2.5$ m, $\Delta z = 0.125$ m; RAM at $(5, 0.25)$ m differs from it by 0.11–0.14 dB (median). With a 70° aperture (table 3, fig. 7) the terrain-following solution differs from the RAM reference by 0.68 and 0.74 dB (median at 50 and 500 m); the stair-step bottom gives 0.42 and 0.59 dB.

The difference is not a discretisation error:

- halving Δx changes it by 0.03 dB (0.69/0.74 → 0.65/0.71 dB);

Table 3: Bermuda pedestal: median / 90th percentile of $|\Delta\mathrm{TL}|$ (dB) with respect to RAM ($\Delta r = 2.5$, $\Delta z = 0.125\,\mathrm{m}$), $r > 500\,\mathrm{m}$.

aperture	solver (grid)	$z = 50\,\mathrm{m}$	$z = 500\,\mathrm{m}$
70°	terrain-following ($\Delta x = 5.0$, $\Delta z = 0.29\,\mathrm{m}$)	0.68 / 2.04	0.74 / 4.54
70°	stair-step ($\Delta x = 5.0$, $\Delta z = 0.67\,\mathrm{m}$)	0.42 / 0.89	0.59 / 2.86
–	RAM ($\Delta r = 5$, $\Delta z = 0.25\,\mathrm{m}$)	0.11 / 0.40	0.14 / 1.04

- raising the flux order to $q = 3$ changes it by less than 0.02 dB (0.66/0.70 dB);
- lowering η_b from 3600 m to 1200 m reduces it to 0.54/0.65 dB.

With $\eta_b = \max B$ the σ-map compresses the 100 m shelf column 36 times, so r jumps by a factor of about 30 across the seabed. Part of the difference is therefore the second-order treatment of this extreme interface row, and a less compressive choice of η_b is preferable for large depth contrasts. The remainder is of the same order as the difference between the two stair-step solvers (ours and RAM). Over a 16° penetrable slope, RAM is itself a stair-step one-way model, and we have no exact reference to say which one-way treatment is closer to the full solution. The exact test of section 10.2 and the wedge of table 2 are the discriminating cases. Times: terrain-following 273 s, stair-step 966 s, RAM 27 s and 97 s.

12 Conclusions and limitations

Orthogonal terrain-following coordinates, built from the tapered harmonic extension of the boundary and its orthogonal trajectories, turn terrain and bathymetry into a generalised density and a weight in the Numerov scheme. The rational propagator, of any order and with the angle limit of the flat scheme on the local grid (theorem 1), and the discrete NLBC (theorem 2) are used without modification. The essential additional ingredient is the flux-conserving step (13). Without it a beam whose angle to the coordinate lines changes loses amplitude.

Limitations and open points.

- *Finite-impedance ground over terrain.* The angle-dependent lower NLBC of [30] assumes a range-independent boundary row; over terrain the boundary coefficients depend on ξ. A local Leontovich condition ($\partial_\eta\psi + ik\varsigma h_\eta\psi = 0$) is straightforward. An exact angle-dependent condition with range-dependent coefficients is open.
- *Boundary smoothing.* The boundary has to be C^2; corners (knife edges, cliffs, vertical walls) are smoothed over the scale σ. Very steep or overhanging relief makes Z_η small and requires a large transition height L.
- *Range-dependent refractivity.* The exterior has to be range-independent, exactly as for the flat NLBC. Inside the transition layer a range-dependent medium is admissible: the tables are per column.
- *Global metric bounds.* The grid is selected with the extreme values of h_ξ and r over the whole domain (corollary 2). Over strongly distorted maps (the Bermuda pedestal: $h_\xi \in [0.87, 1.27]$) this widens the wavenumber interval of the rational approximation considerably and limits the feasible aperture. A range-dependent choice of $\Delta\eta$ is not possible (the grid is fixed), but a local check of the error budget, or a map that minimises the spread of h_ξ, would reduce the pessimism.

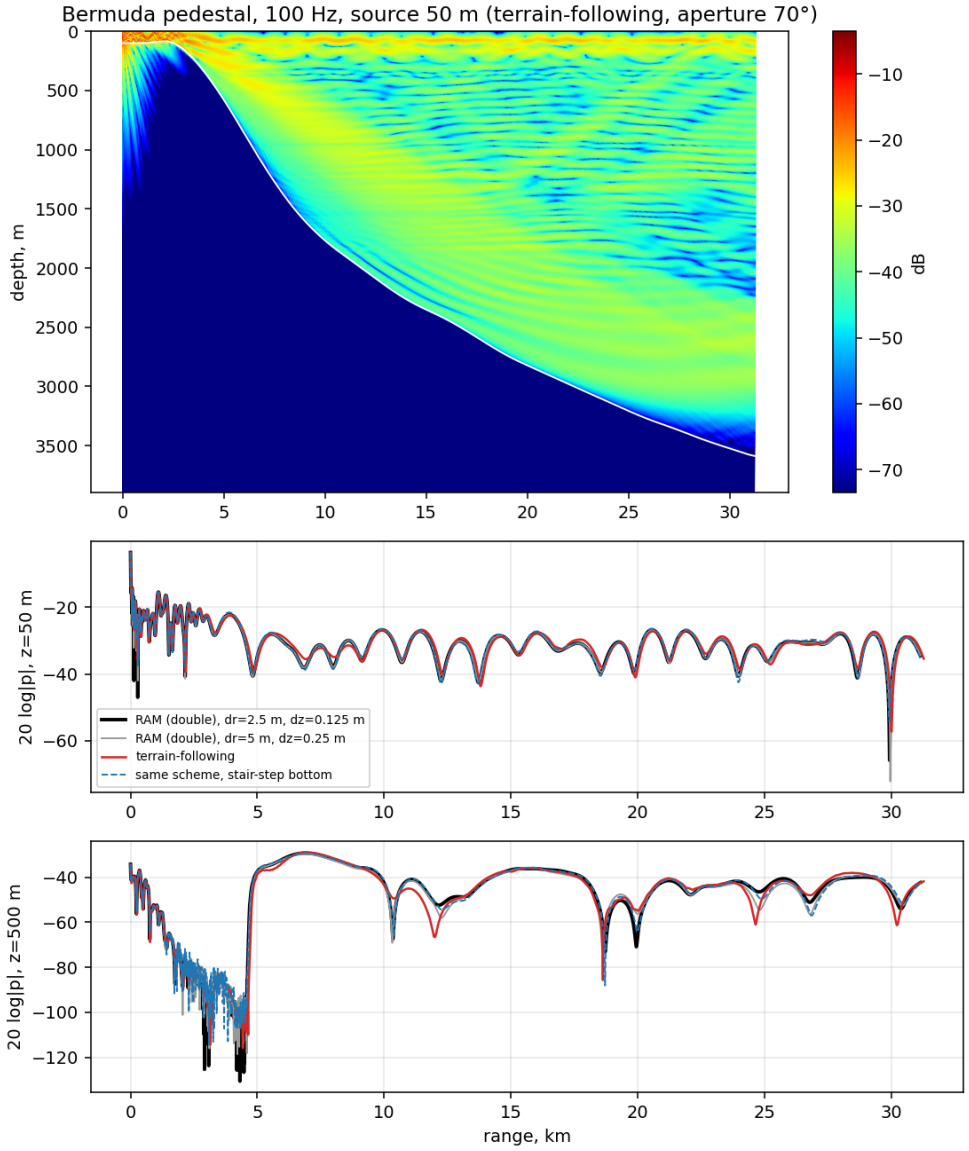


Figure 7: Bermuda pedestal, 100 Hz: field of the terrain-following solver (70° aperture) and $20 \log |p|$ at 50 m and 500 m depth.

- *Large depth contrast in the strip map.* With $\eta_b = \max B$ the σ-coordinates compress a shallow column by $B_{\min}/\max B$. On the Bermuda pedestal this factor is 1/36, so r jumps strongly across the seabed node. A smaller η_b reduces the jump at the price of a larger $\max r$ (finer $\Delta\eta$). A near-conformal profile with guaranteed $Z_\eta > 0$, or a higher-order interface row, would remove the trade-off.

- The starting field is prescribed on the trajectory $\xi = 0$, which is vertical when the boundary is extended evenly at $x = 0$; otherwise it has to be given on a curve.

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